

Zeyuan Hu

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EDUCATION

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|--|------------------------|-----------------------------|
| University of California <ul style="list-style-type: none">• Ph.D. in Computer Science. | Los Angeles, CA | Sept 2025 – Current |
| University of Texas <ul style="list-style-type: none">• Ph.D. studies in Computer Science. | Austin, TX | Sept 2020 – May 2025 |
| University of Texas <ul style="list-style-type: none">• M.S. in Computer Science. (GPA: 3.81/4.00) | Austin, TX | Sept 2017 – May 2019 |
| University of Wisconsin <ul style="list-style-type: none">• B.A. Triple Major, Computer Science, Economics with Honors, Mathematics. (Major GPA Avg: 3.80/4.00) | Madison, WI | Sept 2010 – Dec 2014 |

WORK EXPERIENCE

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| Graduate Researcher <ul style="list-style-type: none">• Formalized CODA [1], a testing framework for Yannakakis’s algorithm family that synthesizes minimal reproducible examples to isolate logical-to-physical translation defects; utilized this framework during the development of TreeTracker Join [2] | UCLA | September 2025 – Current |
| Graduate Researcher <ul style="list-style-type: none">• Developed a provably optimal, empirically fast, and easy-to-implement join algorithm for acyclic conjunctive queries [2]• Co-designed and analyzed two mechanisms for the two-facility location game in a relaxed setting that achieve constant-approximation and constant-strategyproofness, with the second mechanism further guaranteeing Lipschitz continuity [3] | UT-Austin | September 2020 – May 2025 |
| Software Dev Engineer II
AWS Timestream Query Team <ul style="list-style-type: none">• Optimized Hot Tier (HT) query evaluation strategy so that <u>Presto</u> performs intelligent query pushdown and reduced query execution time by 22~29%• Modified Presto parser and optimizer to ensure only selected data types and SQL functions are picked and exposed to the customer• Added end-to-end INTEGER data type support• Combined Timestream internal catalog service with Presto to implement SHOW DATABASES and SHOW TABLES metadata queries• Redesigned the semantics of date & time support in Timestream such that customers can conceptually gauge the correctness of query with date & time string or timestamp | Amazon Web Services | January 2020 – August 2020 |
| Relational Database Intern
Database Infrastructure Group <ul style="list-style-type: none">• Reduced the runtime of long-running queries by over 100× by designing and implementing a query execution feedback technique into ByteNDB, a distributed MySQL modification• Assessed over 80 academic papers to identify promising AI-driven approaches for enhancing cardinality estimation, leveraging statistics collected during query execution• Conducted extensive evaluations on a selected AI-driven approach, running four sets of experiments and collecting over 100 data points to compare its estimation accuracy and inference speed against PostgreSQL, MySQL, and ByteNDB configurations | ByteDance | May 2025 – August 2025 |

Research Intern**Microsoft Research****May 2022 – August 2022**

Database Group

- Formalized a research problem and designed a new algorithm
- Implemented a research prototype to evaluate the algorithm performance in Python on 7 workloads consisting of more than 300 queries

Software Engineer**IBM****August 2015 – August 2017**

DB2 Federation Database Team

- Constructed Hive and Impala wrappers with C++ and Java to support federation database between traditional RDBMS and Hadoop-based data warehouse solution
- Created automated setup tools with Shell that reduce product configuration time by 75%
- Enhanced server option optimization tools using C to reduce federation database performance tuning time by 90 % and enable the capability of tuning the product against Hive, Impala, and Spark
- Resolved over 20 defects, including a severe memory leak issue that impacted a \$1.6 million deal

Cloud Architect Engineer**State Street Financial Service****June 2019 – November 2019**

Omnia Storage Team

- Built auto-deployment system of IBM Cloud Object Storage in multi-site clusters using Ansible and Docker
- Developed a distributed workload generator and performance benchmark toolkit written in Go with Redis, InfluxDB, MongoDB, and Elastic Search

Software Engineer Internship**Schlumberger****May 2018 – August 2018**

HPC Infrastructure Team

- Implemented a monitoring component of the in-house High-Performance Computing (HPC) engine in C++ to provide the fault tolerance and handle the “straggler” problem
- Employed SGD algorithm to dynamically learn the best timing for backup executions of the in-progress tasks based on the computation task characteristics
- Built a C++ code generator that automatically generates the application layer code based on the engine API

Teaching Assistant**UCLA****March 2026 - June 2026**

- Teaching Assistant in Database Systems (Spring 26)

Teaching Assistant**UT-Austin****August 2017 - May 2025**

- Teaching Assistant in Database Systems (Spring 19, Spring 21, Fall 21, Spring 22, Fall 22, Spring 23, Fall 23, Spring 24, Fall 24, Spring 25), Data Engineering (Fall 18, Fall 20), Structure and Implementation of Modern Programming Languages (Summer 21), Quantitative Methods in Neuroscience (Spring 18), Differential Calculus (Fall 17)

HONORS AND AWARDS

- 2025 **Wesley W. Calhoun, Jr. Endowed Scholarship**, UT-Austin
- 2017 **IBM Appreciation program for the Practice: Dare to create original ideas**, IBM
- 2016 **IBM Manager’s Choice Award - Put the Client First**, IBM
- 2018 **Best Internship Project Award (Software Engineering)**, Schlumberger
- 2014 **Graduation with Distinction**, University of Wisconsin
- 2013 **Honors Summer Sophomore Research Apprenticeship**, University of Wisconsin
- 2012 **Meek Bishop Scholarship in Economics**, University of Wisconsin
- 2010-2012 **Dean’s List**, University of Wisconsin

PUBLICATIONS AND PREPRINTS

- [1] Hu, Z. (2026). *Computer-orchestrated design of algorithms: From join specification to implementation*. Retrieved from <https://arxiv.org/pdf/2603.19434>
- [2] Hu, Z., Wang, Y. R., & Miranker, D. P. (2026). TreeTracker Join: Simple, Optimal, Fast. *ACM Transactions on Database Systems*, 51(2).
- [3] Fullerton, E. J., Hu, Z., & Plaxton, C. G. (2025, September). Constant-Approximate and Constant-Strategyproof Two-Facility Location. In *Proceedings of the 18th international symposium on algorithmic game theory* (pp. 119–136).
- [4] Wu, J., Hu, Z., & Mooney, R. (2019, July). Generating Question Relevant Captions to Aid Visual Question Answering. In *Proceedings of the 57th annual meeting of the association for computational linguistics* (pp. 3585–3594).

TALKS AND POSTERS

- Hu, Z. (2025, December). TreeTracker Join: Simple, Optimal, Fast. University of California, Los Angeles. Los Angeles, California, USA.
- Hu, Z. (2025, June). TreeTracker Join: Simple, Optimal, Fast. ByteDance. San Jose, California, USA.
- Hu, Z. (2025, May). TreeTracker Join: Simple, Optimal, Fast. Poster presented at [Southern California Database Day 2025](#). Los Angeles, California, USA.
- Hu, Z. (2024, December). TreeTracker Join: Simple, Optimal, Fast. University of Washington-Seattle Database Seminar. Washington, USA.

SELECTED PROJECTS & RESEARCH EXPERIENCE

- **TreeTracker Join** (2020 - 2025). Implemented a query engine using TreeTracker Join as the core join algorithm; the engine outperforms DuckDB. [Java](#), [Rust](#)
- **RustFS** (2018 - 2019). Building a user-space file system that leverages NVMe SSD. [Rust](#), [SPDK](#)
- **Strata with Lease** (2018). Extended Strata file system with Lease mechanism to support concurrent file access across processes. [C](#)
- **HyperPebblesDB** (2018). Constructed a key-value store that is part of LevelDB family with focus on reducing write amplification. [C++](#), [CMake](#), [Autotools](#)
- **Distributed Key-Value Store** (2018). Built a distributed key-value store with [Python](#) that uses *eventually consistency* model with two session guarantees: *Read Your Writes* and *Monotonic Reads*.
- **SGX Benchmark** (2018). Used a cloud service benchmark (*CloudSuite*) to measure the performance penalty brought by SGX on the IPC between a web server (Nginx) and a PHP application in a local environment
- **Benchmark Journaling Write Amplification** (2018). Measured journaling impact on the write amplification of various file systems using customized [filebench](#), [blktrace](#), [iostat](#), and [strace](#)
- **Benchmark Write Amplification on Fragmented File Systems** (2018). Measured the write amplification of workloads from [filebench](#) and Git workload from *BetrFS* on fragmented (i.e., *aged*) file systems, which are created from file system aging tool (e.g., *Geriatric*)

SPECIALIZED SKILLS

- **Languages:** C++, C, Python, Go, Rust, Shell, SQL, Java, Lisp, MATLAB
- **Software:** CMake, Autotools, QEMU, Docker, Tensorflow, Keras, Git, ClearCase, Hive, Impala, Hadoop
- **Graduate Coursework:** Machine Learning, Structured Models for NLP, Human Computation & Crowdsourcing, Natural Language Processing, Semantics, Distributed Systems, Advanced Operating Systems, Data Centers, Algorithms, Automated Logic Reasoning,

SERVICE AND SOCIETIES

- IBM Diamond & Ring Toastmaster Club (June 2016 - June 2017), President
- UTCS Master Admission Committee (2018, 2021), Member
- UTCS Directed Research Program (DiRP) (September 2020 - December 2021), Mentor